GEA1000 Quantitative Reasoning with Data

Group Project

AY2025/2026 Semester 2

Admin

There are three parts to the GEA1000 Project.

Part A – Critical thinking with data

Part B – Analysing data

Part C – Communicating with data

Parts A and B will be delivered in the form of a Report. Part C will be delivered in the form of a presentation. There will be a question-and-answer (Q&A) session after the presentation which can cover questions from all three Parts of the Project.

Report format

1. Answer all relevant questions from Parts A and B for the Report. Include your answers with reference to the question numbers. You are not required to repeat the questions in your Report.
2. Include sufficient details on data analysis for each question in Part B as an Appendix. These details can be steps or screenshots of your analysis with the software of your choice and should be sufficiently informative for someone else to be able to replicate what you did.
3. Produce a single Report of no more than 10 pages. The cover page, references, appendices, or annexes, if any, do not count towards the page limit. As a guide, Part A should not be more than 3 pages long.
4. Use either Arial or Calibri at font size 11. Set layout as A4 with normal margins of 2.54 cm.

Presentation format

1. Part C will be presented to your tutor in a presentation lasting no more than 10 minutes. This will be followed by a Q&A session lasting 10 minutes.
2. The presentation and Q&A sessions will be conducted in your fifth tutorial session.
3. The Q&A session can cover questions from all three Parts of the Project.

4. There is no restriction on presenter line-up. That is, not everyone in the group is required to present.
5. You should use a set of Slides to aid your presentation. There is no limit to the number of slides you choose to use. You will have to submit the set of Slides following the format given in the next section.

Submission

There are two items for submission: Report and Slides.

1. Nominate two different students from your group to submit an item each to the Canvas submission folder designated for your project group.
2. Both items should be submitted in PDF format. Each item should not exceed 10 MB.
3. Name your Report and Slides in the following format:
    - Report: **[tutorial group code]-[project group number]-report.pdf**
    - Slides: **[tutorial group code]-[project group number]-slides.pdf**
    - For instance, if you are in Project Group 4 of Tutorial Group D01, your submissions should be named as **D01-4-report.pdf** and **D01-4-slides.pdf**.
4. Ensure that the full names and matriculation numbers of all group members are included in both submission items.
5. All submissions will be subjected to plagiarism checks.
6. The schedule of submissions is given in Table 2.

Acknowledging use of AI

You should acknowledge the use of AI if you have completed any work with the aid of an AI tool, by filling up Table 1 and including it in the Appendix of your Report. Note that any submission found to have unattributed usage of AI will be considered as an act of plagiarism, and will be subjected to the corresponding disciplinary actions.

Table 1: Declaration of AI use

AI Tool Used	Prompt and output	How the output is used in the assignment

If an AI tool was used to generate an extensive set of intermediate outputs, you may preserve the relevant sections of the entire transcript and submit it as an Annex of your Report.

Peer- and Self-Evaluation

The Peer Evaluation seeks to qualify your experiences working as a Project group and serves as a guardrail against unfair contributions to the Project. An online evaluation form will be available to all students on Canvas after all presentations have concluded.

1. The Peer Evaluation may affect the final project score of each student.
2. You may submit a peer- and self-evaluation at the conclusion of the project.
3. You may abstain from submitting an Evaluation. In that case, your decision will have no bearing on the group evaluation.

Table 2: Timeline for Project

Week	**Odd week groups**	**Even week groups and NUSC students**
5	Release of Project materials	
5–10	Project work	
10	Online submission of Report & Slides. Deadline: Friday 27 March, 1200hrs	
11	Presentation using submitted presentation Slides and Q&A during tutorial session	Online submission of Report & Slides. Deadline: Friday 3 April, 1200hrs
12		Presentation using submitted presentation Slides and Q&A during tutorial session
Reading	Peer- and self-evaluation submission open: Friday 17 April, 0000hrs – Thursday 23 April, 2359hrs	

Project Brief

“Health is wealth” - a famous proverb which means that being healthy is one of the most valuable assets a person can have. The World Health Organisation (WHO) defines "health" as complete physical, mental and social well-being, rather than simply the absence of disease or illness.

This project invites you to explore the many aspects of health. In this project, you and your groupmates assume the role of interns at NUS Health & Wellbeing, Office of the President at the National University of Singapore (NUS). Your group is tasked with crafting and presenting a data story that can guide a policy to improve health behaviours among NUS students. Details about this are given in Part C of this document.

To aid you in your task, you will be performing background research such as literature reviews and quantitative analysis. In Part A, you will review and critique one article about health: what did you learn and to what extent can the lessons learnt help you in your task? In Part B, you will perform exploratory data analysis on a data set of responses to a health behavioural survey to uncover insights that could form the basis of your data story. Finally, in Part C, you will translate what you have learnt in Parts A and/or B to craft and present a data story that can guide a policy to improve health behaviour.

In this project, we look for evidence that you are able to apply data concepts and do not expect you to possess expert knowledge on demographics, public health or public policy.

Part A

In this section, you will evaluate a research study paper (choose 1 of the 5 papers uploaded into the Canvas folder for Part A) pertaining to a particular aspect of health. Answer all 6 questions in your Report.

While answering the following questions, you should incorporate elements of critical thinking and analysis in your answers, and to discuss their implications (where relevant). We look for evidence that you can apply the concepts covered in the course, and do not expect you to possess domain knowledge of the article’s topic to list all possible implications.

Section 1

1. What was the aim of the study?
2. State the main independent and dependent variables and describe how they were measured.

3. a) Summarise the main finding(s) of the study in one sentence. You may choose to quote a sentence directly from the journal article.
   b) What do you think was the target population for the study?
   c) If a main finding of the study involved a test of hypothesis, what were the null and alternative hypotheses for that test? (If there were multiple tests, choose 1 of them.) Did the authors have sufficient evidence to reject the null hypothesis?

Section 2

4. Answer **one** subsection applicable to your article of choice.

   If the study was a controlled experiment:
   - How were the subjects assigned to treatment and control groups?
   - Was treatment assignment blind to the subjects? How about the assessors?
   - Did the authors of the study report the baseline characteristics of the control and treatment groups? If they did, what do you think was the purpose?

   If the study was an observational study:
   - What potential confounders were controlled for by the researchers?
   - For any one of the variables stated, explain how this variable may be a potential confounder.
   - Is there a potential confounder which was not controlled for? Explain how it may be a potential confounder.

5. Describe the selection process of subjects used by the researchers. Additionally, include answers to the following, or explain why it is not possible to do so:
   - Who were the subjects studied, and how many subjects were there?
   - To what extent did the sampling frame cover the target population?
   - Was probability sampling employed?
   - What was the response rate of the study?

   Based on your answer above, evaluate the generalisability of the study's findings to the target population.

6. Conclude your evaluation of the study as a whole: to what extent was it well-conducted? In your response, you may reflect on the following:
   - Besides its generalisability, what are the strengths and weaknesses of the study? How do they aid/compromise the study's conclusion? How do they weigh against one another?
   - How relevant is the study? Did it provide new insights to the article's topic?

Part B

In this section, your intern group is given a subset of responses to a health behavioural survey done by university students. You will utilise and explore the data set given to generate insights and guide decisions for the crafting of your policy. Download the data set assigned to your group from Canvas. For example, if you are in Project Group 2, you will use "Group2.csv". Details about the data set are presented in the Appendix.

The following sections provide a basic framework for you to conduct exploratory data analysis. Work through the sections and present your responses in your Report.

Section 1: Understanding Data and Data Cleaning

Before running any data analyses, it is good practice to understand your data first. This means one needs to get a sense of what types of variables are present, as well as to prepare the data by cleaning invalid or irrelevant entries.

1. State the number of observations in the data set. For all variables in the data set that you will use in your analysis in Parts B and C, identify if they are categorical, numerical or neither.

We next investigate the data set to determine if data cleaning is required. This involves searching for missing data points, wrongly recorded data, or anomalies.

2. Do a quick scan of the data set and point out any interesting/unusual observations that could affect any subsequent data analysis. How would you deal with this data?

Note: For the subsequent questions, indicate clearly any data cleaning you have performed for each question.

Section 2: Data Visualisation and Analysis

The next step is to explore variables in the data set and their relationships. Summary statistics and visualisations in the form of plots can provide a broad overview of the distributions of the variables. Contingency tables and scatter plots can also help with the analysis of the associations between variables.

One of the ways to achieve a healthier body and mind is to be more physically active. Regular physical activity can help improve energy levels, promote mental health, maintain a healthy

weight, and reduce risks of chronic diseases, among many other health benefits. The WHO recommends adults to engage in at least 150-300 minutes of moderate-intensity aerobic activity per week for substantial health benefits[1].

3. For the variable `PA_Days`,
   (a) Plot an appropriate visualisation to show the distribution of the variable, and
   (b) Compute the five-number summary as well as the mean and standard deviation for `PA_Days`.

   What does this data say about the physical activity of participants?

The Body Mass Index (BMI) is one of the many useful tools we can use to track our health. It measures the ratio of your height to your weight to estimate the amount of body fat you have.

4. Transform the variable `BMI` into a categorical variable `BMI_Class` by classifying each `BMI` value x according to the following Asian BMI cut-off points[2]:

$$x < 18.5 : \text{Underweight}$$
$$18.5 \leq x < 23 : \text{Normal}$$
$$23 \leq x < 27.5 : \text{Overweight}$$
$$x \geq 27.5 \ \text{Obese}$$

   Compare the proportions of each `BMI_Class` across countries.

Regular physical activity can help maintain a healthy body weight by getting rid of excess body fat. As we exercise, our body burns calories, thus burning off our body fat.

5. (a) With an appropriate visualisation, analyse the relationship between time spent daily on physical activity and `BMI`. Choose an appropriate way, using variable(s) in the data set, to quantify time spent daily on physical activity. Justify your choice.

   (b) Do the same as what was done for part (a), but for each of the respective countries. Is there any difference when comparing between countries; as well as comparing with the findings for part (a)? Give a brief description of your observation.

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[1] (2024, June 26). *Physical Activity*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/physical-activity

[2] (n.d.). *Obesity*. Woodlands Hospital. https://www.wh.com.sg/ForPatientsVisitors/conditionstreatments/Pages/Obesity.aspx

Regular physical activity not only improves your physical health, but it also elevates your mood. It is known that exercise can act as a stress reliever by reducing stress hormones and thus fostering relaxation[3].

6. Your friend Dean claims that while a higher level of physical activity is positively associated to relaxedness, gender is a confounder in the relationship between the two variables. Investigate Dean's claim within the data set with the help of the following:
   - Choose a variable in the data set that you think best represents levels of physical activity, justifying your choice. Recode your chosen variable, using a reasonable classification, such that it is a binary categorical variable with values "High PA" and "Low PA".
   - Recode the variable `Relaxedness`, using a reasonable classification, such that it is a binary categorical variable with values "Relaxed" and "Not Relaxed".

Besides regular physical activity, good diet and nutrition is important in keeping our bodies healthy. Eating enough fruits and vegetables every day is one way we can try to achieve a healthier diet.

7. According to the Health Promotion Board, it is recommended to eat 2 servings of fruit and 2 servings of vegetables per day[4]. Your same friend Dean suspects that more than half of **all Singaporean university students** consume less than the recommended servings of fruits per day. Using the variable `Fruits`, conduct a suitable test to test Dean's hypothesis. Comment on whether the result validates Dean's suspicion, considering the limitations or assumptions of the sample.

Section 3: Thinking with Data and Reflecting on Data

At this stage, you would have explored a few variables and made use of data to ask and answer questions. Let's dive deeper for a second round of data analysis!

8. Perform an additional data analysis that builds on what you have done so far. You may choose to explore new variables, ideas or perspectives, so long as you do not repeat exactly what you have done for any of the previous questions.

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[3] (2025, April 23). *How Exercise Balances Cortisol Levels*. Standford | Lifestyle Medicine. https://lifestylemedicine.stanford.edu/how-exercise-balances-cortisol-levels/

[4] (n.d.). *Nutritious Foods For A Healthy Diet*. HealthHub. https://www.healthhub.sg/programmes/nutrition-hub/eat-more

Ultimately, we would like to synthesise useful insights and evaluate our data analysis thus far, to cap off our learning from data.

9. Suppose that you were able to reconduct this survey. If you were able to collect one more variable (beyond this data set) that could be useful in studying health behaviour amongst university students, what would it be, and why? Explain how you would measure that variable, and any considerations you may have when interpreting information from that variable.

10. Evaluate 2 aspects of the data analysis that you have done in Part B. You may choose to discuss the strengths (how your analysis value-adds to understanding the topic of interest) and/or the limitations (how your analysis affects the generalisability of the findings).

Part C

In Part A, you evaluated a research study paper pertaining to a particular aspect of health; in Part B, you performed quantitative analysis on some aspects of health. In Part C, you are to combine what you have learnt into a practical endeavour!

As interns at NUS Health & Wellbeing (HWB), your group is tasked to craft and present a **data story using insights gained from Part A and/or the data set in Part B** that can **guide a policy to improve health behaviour among NUS** students. In crafting your data story, you are encouraged to explore the data set beyond what you have already done in the Part B questions. You will deliver your ten-minute presentation to the programme management team of NUS HWB, portrayed by your tutor.

<u>What is a data story</u>

A data story is a narrative that combines numbers, visuals and real-world context to highlight meaningful insights and inspire action. Data storytelling builds on raw data to highlight relationships, patterns, and implications. It weaves insights into a clear narrative that explains why certain trends matter, and how they should inform decisions.

<u>Elements of a data story</u>

There are three core elements to a data story:

1. Narrative

The narrative is the framework that guides your audience through a series of insights. Think of it in three main parts - beginning, middle, and end - just like most fictional stories.

a. *Beginning: Identify a problem or a question your audience cares about.*
   In the context of this project, identify a health behaviour problem. You should elaborate more about the health behaviour problem identified. Why did you choose that health behaviour problem? How is it a problem? Why should it matter to the audience?
b. *Middle: Show how you analysed the data.*
   This is where your work from Part B comes in. Incorporate data insights and visualisations that relate to the identified health behaviour problem. You can also share relevant insights gained from Part A in this section.
c. *Ending: Present your conclusion and a clear course of action.*
   This is where you propose a policy inspired by the insights gained in Part A and/or B, with the aim to improve the identified health behaviour problem. You should give a short overview of the proposed policy – what is it, how it works and briefly evaluate the strengths and limitations of the proposed policy, and suggest how to overcome them.

The figure below visually summarises how you can structure the narrative of your data story.

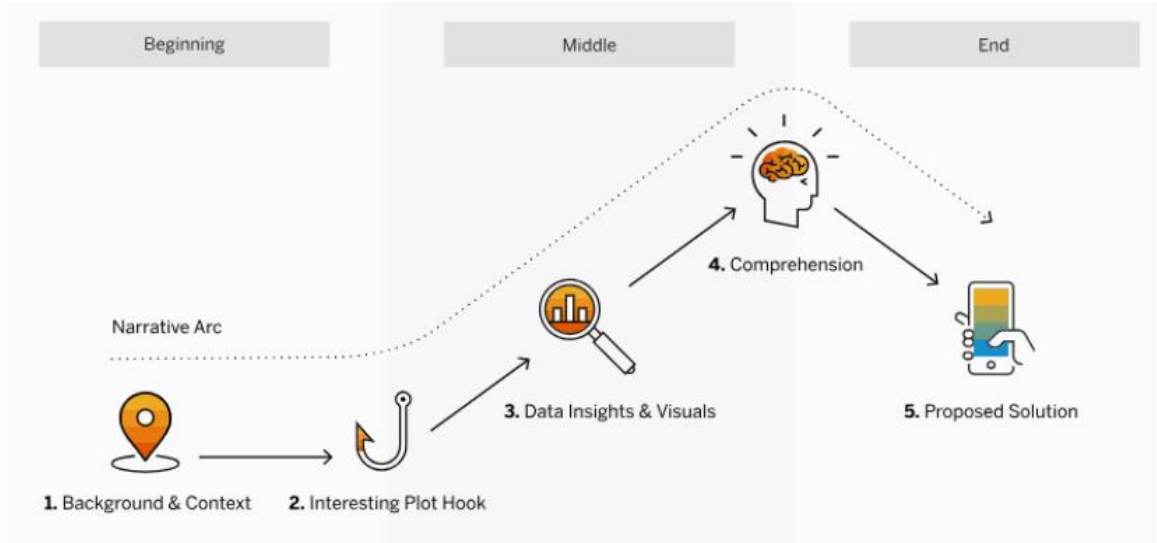

Figure obtained from *What is data storytelling? | Definition, Importance, Examples | SAP.*

2. Data
   As its name suggests, you will need data to create a data story. You will be using only the data set provided in Part B. After analysing the data set, you would have gained

insights which ultimately form the basis of your data story. Besides the analyses that you have already done in Part B, you are encouraged to carry out other analyses that can help make your data storytelling more compelling. You can also incorporate relevant insights gained from Part A in your data story.

3. Visualisations
   Data visualisations not only help make your data story more interesting to your audience, but they are also useful to further explain and uncover data insights. Besides using visualisations that you have already created in Part B, you are encouraged to explore other forms of data visualisations not done in Part B. You may also opt to use data visualisation techniques beyond the scope of what was taught in classes.

You are to use a set of Slides to present your data story. However, should you think that incorporating an Excel dashboard, in addition to your Slides, will aid the presentation of your data story, you may choose to do so. If you do so, you are **not** to submit the Excel dashboard, but you should include screenshot(s) of your dashboard in the Appendix of your Report.

Your presentation will be graded on the following:

1. Data communication: Communicate a coherent and succinct data story to the audience. Use relevant insights and appropriate data visualisations that directly support the narrative.
2. Competency: Propose a relevant and feasible policy backed by evidence. Demonstrate awareness of gaps, limitations or bias of the data story and the proposed policy that were crafted and inspired using insights gained from Parts A and/or Part B, and how to overcome them.
3. Delivery: Excite and engage the audience in a manner that encourages retention of your presentation. Tailor your presentation to the audience.

<u>References and resources</u>

The following are references and resources you may find useful in learning more about data storytelling and Excel dashboarding. Some of these references were also used in the above explanation of data storytelling.

1. Samples of Data Storytelling uploaded on Canvas > Files > Project > Part C > Data Storytelling Samples
2. (n.d.). *What is data storytelling?* SAP. https://www.sap.com/products/data-cloud/cloud-analytics/what-is-data-storytelling.html

3. Strahl, A. (2025, March 11). *What Makes a Data Story?* Dataquest. https://www.dataquest.io/blog/what-makes-a-data-story/
4. [Kenji Explains]. (2022, May 1). *Build an Interactive Excel Dashboard From Scratch* [Video]. YouTube. https://www.youtube.com/watch?v=Q-zO7zetdX8

Appendix

Variables in data set

Variable Name	Description/Question	Codes
ID	Anonymous participant identity number	
Country	In which country were you born?	
PA_Days	In the past week, on how many days have you done a total of 30 minutes or more of physical activity, which was enough to raise your breathing rate. This may include sport, exercise, and brisk walking or cycling for recreation or to get to and from places, but should not include housework or physical activity that may be part of your studying.	0 None 1 1 day 2 2 days 3 3 days 4 4 days 5 5 days 6 6 days 7 7 days
VA_Weekly_Days	In a typical week, on how many days do you do vigorous-intensity sports, fitness or recreational activities?	0 None 1 1 day 2 2 days 3 3 days 4 4 days 5 5 days 6 6 days 7 7 days
VA_Daily_Hours	How much time do you spend doing vigorous-intensity sports, fitness or recreational activities on a typical day (hours)?	
MIA_Weekly_Days	In a typical week, on how many days do you do moderate-intensity sports, fitness or recreational activities?	0 None 1 1 day 2 2 days 3 3 days 4 4 days 5 5 days 6 6 days 7 7 days
MIA_Daily_Hours	How much time do you spend doing moderate-intensity sports, fitness or recreational (leisure) activities on a typical day (hours)?	

Sedentary_Daily_ Hours	How much time do you usually spend sitting or reclining on a typical day (hours)?	
Hours_Slept	Total number of hours slept in a typical day	
Uni_Facilities	Do you use any sports and/or recreation facilities in your university for physical activity? For e.g., gyms, swimming pools, ball courts, sports fields, etc.	Yes (1) No (2) My university does not have these facilities (3) Don't know/Not sure (4)
Smoking_Habits	Which of the following best describes your cigarette smoking status?	I smoke daily (1) I smoke occasionally (2) I don't smoke now, but I used to (3) I've tried smoking a few times, but never smoked regularly (4) I've never smoked (5)
Alcohol_Drinks	On a day when you drink alcohol, how many standard drinks do you usually have? (a standard drink is equivalent to a midi of beer, a glass of wine or a nip of spirits).	None/ Do not drink (0) 1 (1) 2 (2) 3 (3) 4 (4) 5 (5) 6 (6) 7 (7) 8 (8) 9 (9) 10 (10) 11 or more (11)
Fruits	How many serves of fruit do you usually eat each day? (a serve of fruit is equal to one medium piece of fruit or one cup of diced fruit).	1 (1) 2 (2) 3 (3) 4 (4) 5 (5) 6 (6) 7 (7) 8 or more (8)
Veg	How many serves of vegetables do you usually eat each day? (a serve of vegetables is equal to half a cup of cooked vegetables or 1 cup of salad).	

Fastfood	On how many days in the last week did you eat meals or snacks such as burgers, pizza, fried chicken or chips from places like McDonalds, Hungry Jacks, Pizza Hut or Red Rooster?	None (1) 1 day (2) 2 days (3) 3 days (4) 4 days (5) 5 days (6) 6 days (7) 7 days (8)
Sugary_Drinks	How many days per week do you usually drink non-alcoholic sugar-sweetened beverages (such as soda, fruit flavored drink, sweetened milk (with added sugar)?	None/ Do not drink (1) 1 day (2) 2 days (3) 3 days (4) 4 days (5) 5 days (6) 6 days (7) 7 days (8)
Optimism	Over the last 2 weeks, I've been feeling optimistic about the future.	None of the time (1) Rarely (2) Some of the time (3) Often (4) All of the time (5)
Relaxedness	Over the last 2 weeks, I've been feeling relaxed.	
Coping	Over the last 2 weeks, I've been dealing with problems well.	
Closeness	Over the last 2 weeks, I've been feeling close to other people.	
University	Name of university currently enrolled	
Year	What is your current year of study?	Year 1 (1) Year 2 (2) Year 3 (3) Year 4 (4) Year 5 or more (5)
Gender	What is your gender?	Male (1) Female (2)
BMI	Body Mass Index (calculated from weight in kg/ (height in m)^2)	
GPA	Standardised Grade Point Average (GPA) (5-point score)	
Campus	Where do you live?	On campus (1) Off campus (2)
Sports_Club	Did you belong to any sport clubs over the past 6 months?	Yes (1) No (2)

Data set source and collection methodology

The original data set comes from the research paper *University students' overall and domain-specific physical activity during COVID-19: A cross-sectional study in seven ASEAN countries*

The following was taken from *Section 2.1 Study design and participants* of the paper:

This study is part of a larger multinational cross-sectional research project, called "AUN–HPN health behavioral survey". All AUN–HPN member universities were invited to participate in the research project, and their participation was based on a voluntary basis. Eventually, 17 member universities from seven ASEAN countries, namely Brunei Darussalam, Indonesia, Malaysia, Philippines, Singapore, Thailand, and Vietnam accepted the invitation (Table 1). Participants were undergraduate students aged 18–22 years. All eligible undergraduate students in participating universities were invited to complete the online survey. Student recruitment and the online survey distribution were coordinated by representatives of each university and was tailored to the culture and practices of each university. Students were invited to participate through several media, including public relations posters, university-wide email circulation, official university social media channels such as group LINE (Line Corporation, Tokyo, Japan), Instagram, Facebook, and WhatsApp (WhatsApp LLC, CA, USA). Participants accessed the survey through a link or a QR code provided in the media. Students were encouraged to participate in the study by being eligible for a prize draw for fifty smart watches. All participating universities collected data during March to June 2021, when movement and social gathering restrictions were imposed in different forms in all participating universities due to the coronavirus disease 2019 (COVID-19) pandemic.

The final sample obtained in the study consists of 164 variables and 15,366 responses from the seven countries. To simplify the data set for the purpose of Project Part B, the GEA1000 team selected responses from three countries (Singapore, Malaysia and Indonesia) and a subset of variables, while also renaming and recoding some of the variables.